

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE:CSA1703

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.*
(BTL 5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

*I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K. Pujitha

Annexure A

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. **Utility-Based Agent for E-Commerce Recommendations**

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. **Learning Agent for Personalized Education System**

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. **Multi-Agent System for Disaster Management**

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood

Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Reactive Agent-Based Traffic Control System



1. Introduction

A Reactive Agent-Based Traffic Control System is an intelligent traffic management system that uses agents to observe traffic conditions and immediately take suitable actions. Unlike traditional traffic signals that operate using fixed timings, reactive agents continuously monitor the environment and respond to changes such as traffic congestion, accidents, emergency vehicles, and pedestrian movement.

The main objective is to reduce traffic congestion, minimize waiting time, improve road safety, and provide efficient traffic flow.

2. What is a Reactive Agent?

A reactive agent is an intelligent agent that:

- Observes the current environment.
- Receives information through sensors.
- Applies predefined rules.
- Takes an immediate action.
- Does not necessarily maintain a detailed history of previous situations.

For example:

IF traffic density is high on Road A
THEN increase the green-light duration for Road A.

3. System Components

Sensors

- CCTV cameras
- Vehicle detectors
- Infrared sensors
- GPS information
- Emergency vehicle detection systems

Agent

- Receives traffic information.
- Determines the current traffic condition.
- Applies decision rules.
- Sends commands to traffic signals.

Traffic Signal Controller

- Controls red, yellow, and green signals.

- Changes signal timing according to the agent's decision.

Communication Network

- Transfers information between intersections and traffic-control agents.

Instead of traffic signals changing only according to a fixed timer:

Sensor detects traffic → Agent analyzes it → Signal reacts

Basic System Architecture

Traffic Cameras / Sensors



Traffic Data



Reactive Agents



Traffic Condition Analysis



Decision Making



Traffic Signals



Traffic Flow Improvement



New Traffic Data